



Two dynamic energy budget models for the harpacticoid copepod *Nitocra spinipes*

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ABSTRACT

The harpacticoid copepod *Nitocra spinipes* is a commonly used test species in ecotoxicological studies and subject of multiple international testing guidelines. While we strive for a better understanding of toxicant-induced effects in this species, physiological models rooted in the Dynamic Energy Budget (DEB) theory can help to explain such effects on the level of energy allocations within the organism. In previous works that aimed to capture the copepod life history in a DEB model, diverging assumptions on the growth pattern of copepods were made. While some authors presumed von Bertalanffy growth, others presumed an upcurving in length versus time due to metabolic acceleration. In this study we parametrized the two typified DEB models “abp” (metabolic acceleration from birth to puberty) and “sbp” (standard von Bertalanffy growth from birth to puberty) on life history data of *N. spinipes*. Besides using data from literature, we also measured additional length-at-time data to aid the parameter estimation. As the body proportions of *N. spinipes* changed continuously throughout its development, we used the square root of the top view area as a length measure to scale with the cube root of structural volume in length-to-volume conversions. Experimental data were predicted well with both models according to goodness of fit criteria. Despite a slightly better data fit in abp, we cannot rule out sbp as implausible. Overall, we expect both models to perform equally well in future applications. More detailed data on *N. spinipes* and closely related species are needed to support or reject the presumption of metabolic acceleration in the life history of copepods.

1. Introduction

The harpacticoid copepod *Nitocra spinipes* has become a common test species in ecotoxicological studies over the last decades (Bengtsson, 1978; Breitholtz et al., 2008; Ribbenstedt et al., 2017). As a benthic member of mainly coastal and estuarine communities, this species plays an important role as prey for higher trophic levels (Gee, 1989). Since *N. spinipes* is subject of multiple international ecotoxicity testing guidelines (ISO, 1999, 2016; OECD, 2014), there is a growing body of acute and chronic toxicity data for this species. Unfortunately, the mechanisms that lead to observable life history effects in *N. spinipes* are usually unknown. A better understanding of such mechanisms and how they are influenced by variations in environmental conditions such as food quantity and temperature is, however, crucial for proper laboratory-to-field extrapolations. The Dynamic Energy Budget (DEB) theory (Kooijman, 1993, 2010b) holds great explanatory potential for the processes in question. As all animals obtain energy from food, the DEB theory comprises a set of simple energy allocation rules which govern the processes of maintenance, growth, maturation, and reproduction of

an individual while simultaneously respecting the law of conservation of energy. In principle, all toxic effects occurring at the organism level can be explained by stress-induced alterations in these basic DEB processes (Ashauer and Jager, 2018). Model simulations can be tested against experimental toxicity data to help pin down the most likely physiological mode of action of a chemical leading to an observed life history effect.

The simplest conceptualization of an animal's life history in the DEB theory is the ‘standard DEB model’ (Kooijman, 2010b). In this model, the feeding rate, as well as the related assimilation rate of an animal, is proportional to its volume^{2/3}. The assimilated food energy is at first stored in a reserve compartment. The reserve mainly serves as a buffer to smooth out fluctuations in food quantity and quality. For further evidence-based justifications of this model compartment see Kooijman (2010b). A fraction κ of remobilized reserve energy is allocated to the processes of somatic maintenance and structural growth. Maintenance costs are always accounted for first whereas the surplus is invested in growth. Concurrently, the fraction $1 - \kappa$ is allocated to the processes of maturation and maturity maintenance. When puberty is reached,

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investment in maturation ceases and investment in reproduction begins. As with somatic maintenance, maturity maintenance costs are paid first and only the surplus is invested in maturation or reproduction.

Ideally, these DEB model rules would consistently apply to all animals in a uniform manner, with interspecific differences being covered by deviations in the model parameters only. However, in the process of calibrating DEB-based models for a wide range of species from different taxa, slight deviations from the above presented standard DEB model have been found which has led to the introduction of a set of ‘typified models’ (DEBwiki, 2017; Kooijman, 2010a). These models typically include additional life stages during which animals show characteristic behavior in terms of their feeding, growth or reproduction. A common deviation from the standard DEB model is metabolic acceleration which is usually indicated by a period of upcurved growth (in length versus time) in an animal's life history. Kooijman (2014) recently presented an overview of different types of metabolic acceleration. The most common form is the type *M* acceleration in which an animal's surface-area-specific assimilation rate and its energy conductance (which affects reserve mobilization) increase proportionally to its structural length, i.e., volume^{1/3}.

Until now, copepods have mainly been considered to follow type *M* metabolic acceleration from birth until puberty, the latter being induced by their final molt (Kooijman, 2010a). After puberty, growth ends as reproduction is initiated. This characteristic variation of the animal life history is denoted by the model code ‘abp’ which stands for acceleration from birth to puberty. So far, abp models for four different copepods (*Acanthocyclops robustus*, *Cyclops vicinus*, *Mesocyclops leukarti*, and *Eurytemora affinis*) have been parametrized on experimental data and entered to the Add-my-Pet (AmP) data collection (AmP, 2017). Jager et al. (2015) parametrized a simplified reserve-less version of the standard DEB model (DEBKiss; Jager et al., 2013) for the calanoid copepod *Calanus sinicus*. Model modifications capturing the abrupt stop in growth at puberty closely resembled the typified model abp. However, *C. sinicus* was not observed to accelerate. Based on this study, an entry to the AmP collection was made for *C. sinicus* introducing the typified model ‘sbp’ which stands for standard (as in the standard DEB model) metabolic behavior from birth to puberty. Since these approaches to reproduce the copepod life history in a DEB model all delivered reasonable fits to existing data, we consider none of the underlying data sets strong enough to soundly prove or disprove the occurrence of metabolic acceleration in copepods. Therefore, in this study we considered both typified models abp and sbp as possible options to capture the life history of *N. spinipes*.

In this study, it was our aim to parametrize the two typified models abp and sbp on life history data of *N. spinipes* and discuss both models in terms of their plausibility and applicability. Besides using data from literature, we performed a growth experiment to generate data on length over time to aid the parameter estimation and the subsequent model evaluation. As the body proportions of *N. spinipes* continuously changed during its development, we paid special attention to finding a suitable, yet simple length measure scaling with its structural volume^{1/3} to facilitate length-to-volume conversions. This is the first study to provide DEB parameters for *N. spinipes*, an ecotoxicological model species that has never been described in a DEB context before. Beyond that, the methods described in this study may help other researchers to parametrize DEB models for other copepods more easily in the future.

2. Material and methods

2.1. Test organisms

Specimens of *N. spinipes* were kindly provided by the Department of Environmental Science and Analytical Chemistry (ACES) at Stockholm University, where *N. spinipes* has been in continuous culture ever since 1975 when it was isolated from the Tvären Bay, Baltic Sea. A new

culture was established in our laboratory following the methods described by Breitholtz and Bengtsson (2001). Diluted seawater was used as cultivation medium. Natural seawater was obtained 500 m offshore from the coast of Blankenberge, Belgium. The water was filtered with 0.2 µm membrane filters and diluted with deionized water to a salinity of 7‰ before use. The culture was permanently maintained in darkness at 22 °C.

2.2. Growth experiment

While development times in between certain life stages of *N. spinipes* have been reported frequently in the past, high quality data on length over time have, to our knowledge, never been reported. Consequently, in this study we performed a growth experiment with *N. spinipes* over its complete life history up until puberty. The experiment was carried out in the same medium as was used for culturing (see subsection 2.1). Newly hatched nauplii (< 6 h) were transferred to 6-well plates with approximately 50 nauplii in 8 mL of medium per well. The plates were incubated in darkness at 22 °C and were fed with the cryptophytic alga *Rhodomonas salina* (2.5×10^5 cells mL⁻¹). Medium changes (70% renewal) and feedings were performed every first, third, and fifth day of a week. At every length measurement, between 10 and 20 animals were picked randomly from different wells and measured under a light microscope (Olympus SZX2-ILLK; cellSens Dimension 1.13). In addition to an animal's body length (without antennae and cerci), we measured its prosome and urosome widths as well as its top view area as illustrated in Fig. 1. Measurements were performed daily (except for days 11 and 12) until day 17 when most of the sampled animals were in the adult stage. Two additional measurements were performed on the days 22 and 28 after which the experiment was terminated. From day 14 on, the sex of each animal was determined by primary and secondary sexual characteristics as described by Koch et al. (2017).

2.3. Data compiled for the parameter estimation

2.3.1. Development time

Copepods pass through six larval stages (nauplius form) before they undergo a complete metamorphosis to the copepodite form (see Fig. 1). After the complete metamorphosis they molt five more times before they reach the reproducing stage, i.e., ‘puberty’ in DEB terminology. As the events of hatching, metamorphosis, and the final molt can be determined easily under a light microscope, development times between these events have been studied extensively for *N. spinipes* in the past (Dahl and Breitholtz, 2008; Koch et al., 2017; Lundström et al., 2010). In this study's parameter estimation we used mean values of the age at hatching a_h (= inter-brood duration), the time from hatching to metamorphosis t_j (= naupliar development time), and the time from metamorphosis to the final molt t_p (= copepodite development time) based on multiple studies at control conditions of 22 °C and *ad libitum* food intake as presented by Koch et al. (2017). Because the exact time at which an embryo started developing after the previous hatching event could not be determined microscopically, we estimated it as an auxiliary model parameter t_0 .

2.3.2. Reproduction

Once fertilized by a male, adult females can produce six or more consecutive broods which are usually released in periods of two to four days. Data on the mean brood size and inter-brood period of multiple control datasets were extracted from Koch et al. (2017). The mean brood size was divided by the mean inter-brood period to obtain the maximum daily reproduction per female R_i .

2.3.3. Dry weight density

Goodman (1980) measured dry weights of ten different harpacticoid copepod species and presented the linear regression $W = 14.01 L - 5.69$ ($r = 0.92$) where W is the dry weight of an adult male or non-gravid

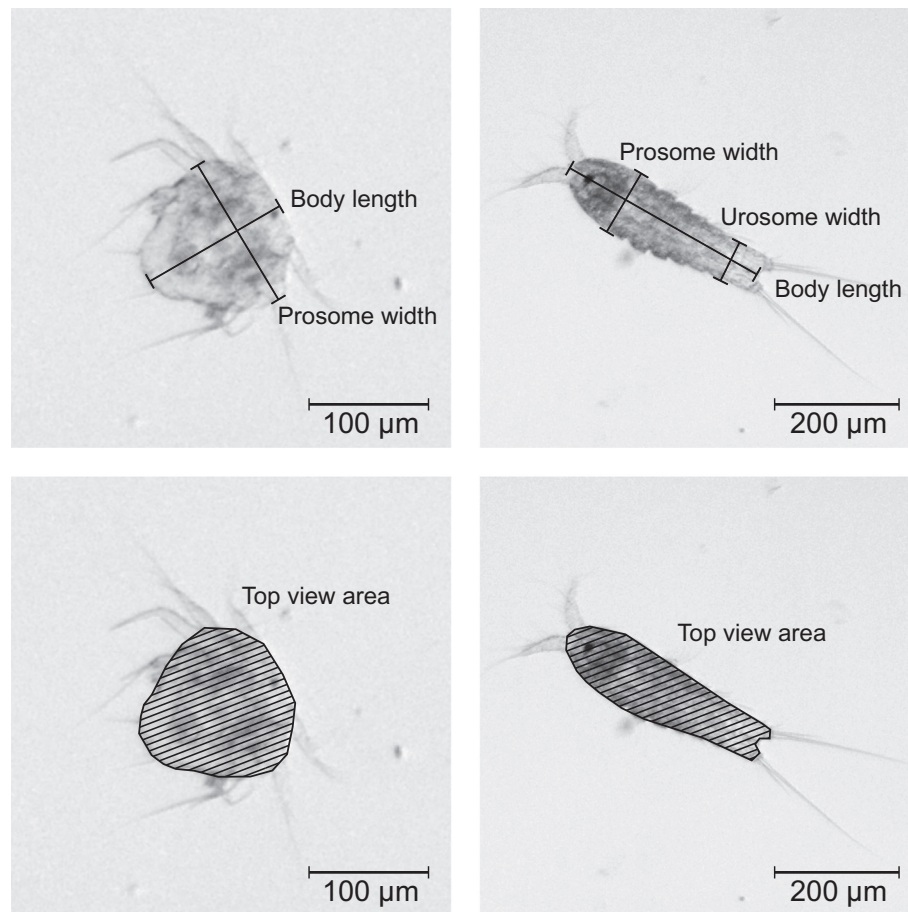


Fig. 1. Body size measures obtained in the growth experiment shown exemplarily for a nauplius (left) and a copepodite (right). The urosome width was measured for copepodites only. Legs, antennae, and cerci were excluded from all length, width, and area measurements.

female in μg and L is its body length in mm. This formula was used to estimate the dry weight of adult females W_d^i of *N. spinipes* based on their average body length as measured in our growth experiment.

2.3.4. Longevity

As no longevity data for *N. spinipes* were available in literature, its maximal lifespan a_m was estimated as the ratio of the maximal lifespan to the hatching-to-adult development time of the related species *C. sinicus* (Uye, 1988) multiplied by the hatching-to-adult development time of *N. spinipes* (all data at 22 °C). While these data are preliminary estimates, they are only relevant to the estimation of the Weibull aging acceleration $\hat{h}_a$ and do not affect any of the more fundamental DEB parameters.

2.3.5. Respiration

The somatic maintenance rate of an animal can typically be captured from the curvature of its asymptotic growth curve (Sousa et al., 2010). Since *N. spinipes* was found to grow rather linearly in length over time and stops growing abruptly after its final molt (see Fig. 3), additional data were required for a realistic estimate of the somatic maintenance rate coefficient $\dot{k}_M$. In the DEB theory, all mass fluxes are weighted sums of the three basic fluxes assimilation, dissipation, and growth. Dissipation here is defined as the sum of somatic and maturity maintenance, maturation, and reproduction overhead costs. As a consequence, data on mass fluxes can help to estimate $\dot{k}_M$ in relation to those parameters which can be estimated well from the other data. Here, we used the average post puberty respiration rate (as O_2 consumption) J_{O^p} of four brackish water harpacticoids in the size range of *N. spinipes* (Herman and Heip, 1983) to aid the parameter estimation.

2.3.6. Length over time

Since the body proportions of *N. spinipes* change continuously throughout its development (see Fig. 2) its body length does not scale stably with its structural length, i.e., volume^{1/3}. Instead, we used the top view area^{1/2} as a more stable length measure to derive volume from length and vice versa (for elaborations see results subsection 3.1). As sexual differentiation in most animals was observed on day 14, only data of females were considered after this point.

2.3.7. Temperature dependency

The effects of temperature on metabolic rates are usually described well by the expressions of Arrhenius (Kooijman, 2010b). The metabolic rate $\dot{k}$ at a given temperature T (in Kelvin) can be written as $\dot{k}(T) = \dot{k}_{REF} \exp(T_A/T_{REF} - T_A/T)$ where $\dot{k}_{REF}$ is the metabolic rate at a reference temperature T_{REF} and T_A is the Arrhenius temperature, a species-specific constant. As the development time of an organism in the DEB theory is inversely proportional to its metabolic rate, this function assumes equiproportionality, i.e., a constant, temperature-independent ratio of development times between all life stages (Corkett, 1984). While many studies support equiproportional development in copepods, deviations have been reported for multiple species including *N. spinipes* (Hart, 1990; Koch et al., 2017; Sarvala, 1979). Nevertheless, with caution against overfitting our models to the data and for reasons of consistency and comparability with other entries in the AmP collection we considered the Arrhenius temperature correction the most suitable for our models. We used data on the inter-brood duration, naupliar development time, and copepodite development time at three different temperatures of 15, 20, and 25 °C (Koch et al., 2017) to determine T_A in the parameter estimation procedure.

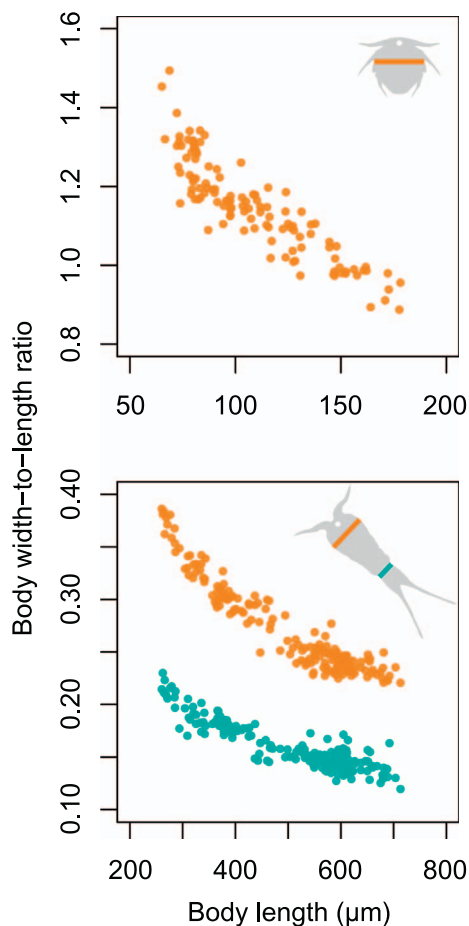


Fig. 2. Body width-to-length ratio versus absolute body length of nauplii (top) and copepodites (bottom) of *N. spinipes*. Orange and blue dots represent the prosoma and urosome widths respectively. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

2.3.8. Food dependency

The feeding rate of an animal can be modeled as a function of food density. If $x = X/K$ is defined as the scaled food density with the absolute food concentration X and the half-saturation coefficient K then the scaled functional response can be written as $f = x/(1 + x)$ (Holling's type II). Data on the naupliar and copepodite development times as well as the female reproduction rate at six different constant algal concentrations of *R. salina* between 1.25×10^4 and 5×10^5 cells mL^{-1} (Koch et al., 2017) were used to estimate K in the parameter estimation procedure. Cell numbers were translated to carbon biomass with a conversion factor of 23 pg C cell $^{-1}$ (Karlson, 2017).

2.4. Typified models abp and sbp

The *N. spinipes* life history data mentioned above were used to independently estimate parameters for the two typified models abp and sbp. While in sbp an animal's surface area scales with its volume $^{2/3}$, in abp it is assumed to scale with volume 1 . Mathematically, this implies that the parameters surface-area-specific maximum assimilation rate $\{\dot{p}_{Am}\}$, surface-area-specific maintenance rate $\{\dot{p}_T\}$, and energy conductance $\dot{v}$ must be multiplied by a shape correction function $M(V) = (V/V_b)^{1/3}$, with V_b being the volume at birth, to derive an abp from an sbp model. As a consequence, abp animals show an upcurving in growth (type M acceleration) compared to the von Bertalanffy growth of sbp animals. In both typified models, puberty induces the end of growth and hence terminates the fractionation of mobilized energy between soma on the one hand and maturation plus maturity

maintenance on the other hand (κ -rule). After puberty, all energy not required for maintenance is invested in reproduction.

2.5. Parameter estimation

We applied the software package DEBtool written in Matlab (Marques et al., 2018) which offers fast and automated estimation routines for DEB parameters. Elaborate guidance on the use of DEBtool can be found via <http://www.bio.vu.nl/thb/deb/deblab/>. In brief, DEBtool makes use of three different files which are called the “mydata” file, the “pars_init” file, and the “predict” file. The mydata file contains all experimental data that are used in the estimation procedure. The pars_init file contains initial values for the model parameters to be estimated. The model functions that are used to predict life history observables from the parameters are formulated in the predict file. The predictions for all experimental data are evaluated simultaneously by means of a loss function. The parameters are adjusted iteratively to minimize the loss function and optimize the model fit.

In this study, we created the DEBtool files separately for the models abp and sbp. All above-mentioned life history data for *N. spinipes* were collected in the mydata files. A set of typical parameter values of a ‘generalized animal’ (Kooijman, 2010b, Table 8.1) was used for initial parameters in the pars_init files. Since not all parameters could be expected to be estimated from the available data, some were fixed at their default values (see Table 1). Model-specific formulae for life history predictions were entered in the predict files. All formulae correspond with the rules of the DEB theory as described by Kooijman (2010a, 2010b). Starting with the initial parameters, we applied the Nelder-Mead method (Nelder and Mead, 1965) to minimize the loss function (multiplicative symmetric bounded). The routine was terminated as soon as the loss function converged to a stable minimum. All DEBtool files used in this study can be found in the data repository linked to this article. Particularly, the predict files can help readers to understand how life history observables were computed from the DEB parameters in each respective model.

2.6. Goodness of fit criteria

The mean relative error (MRE) and the symmetric mean squared error (SMSE) were computed for both models as goodness of fit criteria. On a scale between 0 and infinity, the MRE gives a measure of the additive difference between model predictions and data. The SMSE, by contrast, has values between 0 and 1 and assesses the difference between model predictions and data multiplicatively. In both cases, a value of zero indicates a perfect fit of model predictions and data. The software package AmPtool written in Matlab (Marques et al., 2018) was used to compute percentiles of the goodness of fit values of both models in relation to the full record of entries in the AmP collection (883 entries on December 08, 2017). Furthermore, the median MRE and SMSE, i.e., the 50% percentile of the AmP collection were extracted for reference.

3. Results and discussion

3.1. Growth and shape

A complete overview of the measured body size data can be found in the data repository linked to this article. The body width-to-length ratios versus the absolute body lengths of all measured animals are presented in Fig. 2. The data indicate that the relative body widths of both, nauplii and copepodites, decrease with increasing body lengths. In other words, longitudinal growth in *N. spinipes* is overproportional to its lateral growth and animals grow more slender in the course of their life history. While the body length of an animal is commonly taken to scale with its volume $^{1/3}$ as a first approximation in DEB studies, in case of *N. spinipes* this would lead to an underestimation of its structural volume

Table 1
DEB parameters of the typified models abp and sbp optimized to life history data of *N. spinipes*.

Symbol	Unit	Interpretation	abp	sbp
Fixed model parameters				
κ_R	–	Reproduction efficiency	0.95	0.95
$\{p_T\}$	$J\ d^{-1}\ cm^{-2}$	Surface-area-specific somatic maintenance rate (always zero in ectotherms)	0	0
k_j	d^{-1}	Maturity maintenance rate coefficient	0.002	0.002
T_{REF}	K	Reference temperature	293.15	293.15
Estimated model parameters				
$\{p_{Am}\}$	$J\ d^{-1}\ cm^{-2}$	Surface-area-specific maximum assimilation rate	15.81	48.17
$\dot{v}$	$cm\ d^{-1}$	Energy conductance	0.0206	0.0239
κ	–	Allocation fraction to soma	0.779	0.610
$[p_M]$	$J\ d^{-1}\ cm^{-3}$	Volume-specific somatic maintenance rate	593.30	817.37
$[E_G]$	$J\ cm^{-3}$	Volume-specific costs for structure	4447.90	4454.72
E_H^b	J	Maturity level at birth	5.88×10^{-4}	6.12×10^{-4}
E_H^i	J	Maturity level at metamorphosis	4.10×10^{-3}	1.64×10^{-2}
E_H^p	J	Maturity level at puberty	3.24×10^{-2}	8.18×10^{-2}
$\dot{n}_a$	d^{-2}	Weibull aging acceleration	3.26×10^{-6}	2.66×10^{-6}
δ_M	–	Shape correction coefficient	0.865	0.845
t_0	d	Time at start of development since hatching of previous brood	1.64	2.37
T_A	K	Arrhenius temperature	3062.39	2751.75
K	$\mu g\ C\ mL^{-1}$	Half-saturation coefficient for algal food source <i>R. salina</i>	0.212	0.248
Computed model parameters				
$[E_m]$	$J\ cm^{-3}$	Maximum reserve density: $[E_m] = \{p_{Am}\}/\dot{v}$	766.69	2018.39
g	–	Energy investment ratio: $g = [E_G]/(\kappa \times [E_m])$	7.45	3.62
k_M	d^{-1}	Somatic maintenance rate coefficient: $k_M = [p_M]/[E_G]$	0.133	0.183

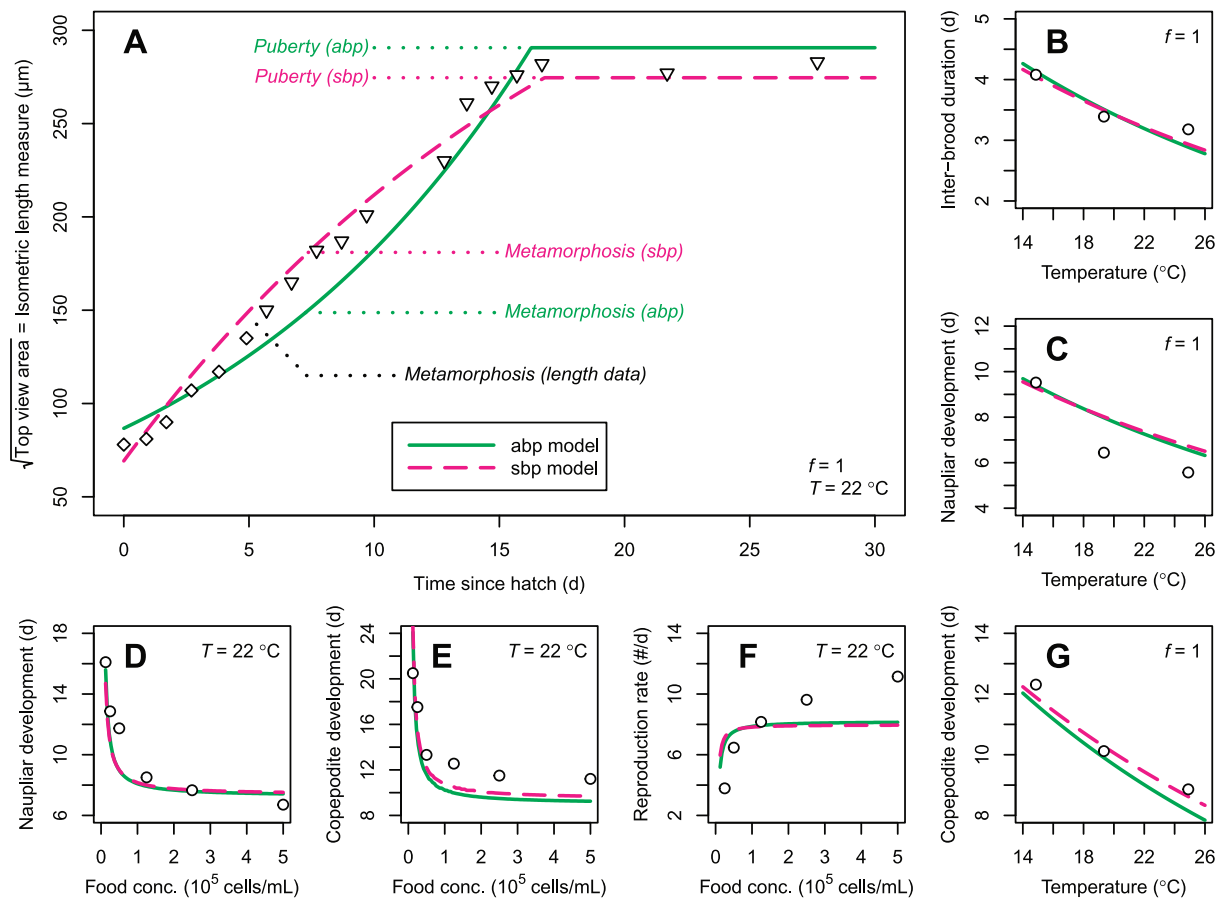


Fig. 3. Univariate life history data and model predictions in the typified models abp and sbp. A shows length over time. Diamonds and triangles represent naupliar and copepodite stages, respectively. B, C, and G show the temperature-dependent data on the inter-brood duration, naupliar development time, and copepodite development time. D, E, and F show the food-dependent data on the naupliar development time, copepodite development time, and reproduction rate.

at early life stages and an overestimation at later life stages, accordingly. In this study, we chose the top view area^{1/2} as a length measure to scale with volume^{1/3}. This length measure is stable to morphological changes in the two dimensions length and width and leaves height as the only dimension to be estimated. The smooth increase of the top view area^{1/2} over time, even at the complete metamorphosis from nauplius to copepodite (see Fig. 3), supports its scalability with structural volume^{1/3} which we also expect to increase smoothly over time.

Based on our results, we suggest that the possibility of morphological changes should always be considered carefully in DEB model parametrizations for new species. Improper length-to-volume extrapolations in animals that defy isomorphy can overestimate growth, and may falsely indicate metabolic acceleration. Unfortunately, detailed morphological data for pre-adult animals are not always easy to obtain. Alternatively, structural volumes can be estimated by means of weight measurements which obviate the need for shape considerations altogether.

3.2. Parameter estimates

All model parameters used in the typified models abp and sbp along with optimal fit values for each respective model are presented in Table 1. While the number of parameters and their denotation is identical between abp and sbp, in abp the parameters $\{\dot{p}_{Am}\}$, $\{\dot{p}_T\}$, and $\dot{v}$ are multiplied by the shape correction function $M(V)$. $\{\dot{p}_T\}$ is of minor importance here as it is always zero for ectotherms. $\{\dot{p}_{Am}\}$ and $\dot{v}$, however, essentially influence the trajectory of the state variables reserve, structural volume, and maturity. They also indirectly control estimates of the other model parameters all of which are connected to the state variables via mathematical functions corresponding with the rules of DEB theory. For a better understanding of the model functions, we provided a hands-on implementation (Excel spreadsheet) of both models abp and sbp in the data repository linked to this article. More details on individual parameters and their interpretation were given by Kooijman (2010b, 2014).

3.3. Model predictions

The goodness of fit criteria of both typified models abp and sbp fitted to the available life history data of *N. spinipes* are presented in Table 2. Both models delivered good fits to the data according to their respective MRE and SMSE. With percentiles between 36 and 45%, the goodness of fit criteria of both models attained values below the median of all entries in the AmP collection. Overall, the fit of abp was slightly better compared to sbp with lower MRE and SMSE values. Model predictions for all zero-variate and univariate data are presented in Table 3 and Fig. 3, respectively.

3.3.1. Zero-variate data predictions

Most zero-variate data were reproduced well by both models with deviations below 10%. The length at hatching L_h was overestimated by 12% in abp, whereas it was underestimated by 12% in sbp. Considering the biological variation in the data, these are still reasonable fits. While the length at metamorphosis L_j was covered well by abp, it was

overestimated by 21% in sbp. This can be explained by the fact that the maturity level at metamorphosis E_H^b was parametrized on zero- and univariate literature data on time between hatching and metamorphosis while the growth curve (Fig. 3A), was fitted to data from our own growth experiment. In the growth experiment, however, metamorphosis was reached about two days earlier compared to the literature data. This slight discrepancy leads to a later metamorphosis in both models abp and sbp compared to the growth experiment. In abp, a later metamorphosis compensates for an underestimated length in this section of the growth curve and results in a good prediction of L_j . In sbp, however, a later metamorphosis results in a much higher L_j as length is slightly overestimated in this section of the growth curve.

3.3.2. Predictions of length-at-time

The length-at-time data were predicted slightly better in sbp compared to abp (Fig. 3A). While abp assumes an upcurving in length over time, no such pattern was visible in the data. In sbp, by contrast, length over time follows an asymptotic growth curve which is truncated at the onset of puberty. If the theoretical positive limit of the (untruncated) growth curve is substantially higher than the length at puberty, the curvature is comparably low between birth and puberty. Hence, this curve type could be fitted well to the experimental data which suppose a rather linear growth pattern.

3.3.3. Predictions of temperature and food dependency

Predictions for the temperature- and food-dependent data on development times and the reproduction rate were almost identical in abp and sbp (Fig. 3B–G). Temperature and food effects were modeled as described in the methods subsections 2.3.7 and 2.3.8 and are independent of the respective model type. As both models showed similar predictions in the zero-variate data of development times and the reproduction rate at standard conditions, similar temperature and food dependency curves of these endpoints could be expected.

While the temperature-dependent data on the inter-brood duration and the copepodite development time were predicted well by both models, the naupliar development time was overestimated at the higher temperatures. As mentioned above, the experimental data by Koch et al. (2017) show slight deviations from equiproportional development. As the expressions by Arrhenius presume equiproportionality, by definition, the data cannot be captured perfectly by our model predictions. Nevertheless, as only a single parameter needed to be estimated, we consider the temperature correction with the Arrhenius temperature T_A as a good trade-off between model complexity and predictive power.

In both models, the food dependency of the naupliar development time was predicted well by means of the functional response f (Fig. 3D). The copepodite development time, however, was underestimated by more than a day at food concentrations above 1.25×10^5 cells mL⁻¹ (Fig. 3E). Again, this can be explained by a slight discrepancy in the data sets used in the parameter estimation. While the zero-variate data point for the copepodite development time t_p is 9 days, and the temperature-dependent data also suggest a value around 9 days at 22 °C (i.e., standard conditions), the copepodite development time in the food-dependent data was above 11 days even at the highest food concentration, which was twice as high as the supposed concentration for *ad libitum* food intake of 2.5×10^5 cells mL⁻¹. As all data were fitted simultaneously, this inconsistency led to a vertical offset between the data and the model predictions. In spite of that, the general curve shape of the model predictions matches the trajectory of the data points well. The food-dependent daily reproduction rate per female (Fig. 3F) was less well predicted by both models. Here again, the univariate data are not perfectly consistent with the zero-variate data (*ad libitum* food intake), which affects the prediction curves especially at high food conditions. Moreover, the curve shape of the model predictions does not match the data trajectory very well. The models predicted the same asymptotic behavior for the reproduction rate as for the development times, whereas the measured reproduction rate did not level off at the

Table 2

Goodness of fit criteria of the typified models abp and sbp fitted to life history data of *N. spinipes*. Median values of the AmP collection are presented for reference.

Goodness of fit criterion	<i>N. spinipes</i> abp	<i>N. spinipes</i> sbp	AmP* median
Mean relative error (MRE)	0.063	0.076	0.083
MRE percentile in AmP*	36%	45%	50%
Symmetric mean squared error (SMSE)	0.071	0.079	0.096
SMSE percentile in AmP*	36%	42%	50%

* Entire AmP collection as of December 08, 2017 (883 entries) accessed via AmPtool.

Table 3Zero-variate life history data of *N. spinipes* and model predictions in the typified models abp and sbp.

Symbol	Unit	Interpretation	Data	Prediction in abp	Prediction in sbp	Source
a_h	d	Age at hatching (counted from the previous hatching)	3.17	3.19	3.21	Koch et al. (2017)
a_m	d	Maximum age	109	109	109	Koch et al. (2017); Uye (1988)
t_j	d	Time since hatching at metamorphosis	7.75	7.26	7.37	Koch et al. (2017)
t_p	d	Time since metamorphosis at puberty	9.00	9.01	9.44	Koch et al. (2017)
L_h	μm	Length* at hatching	78	87	69	This study
L_j	μm	Length* at metamorphosis	150	149	181	This study
L_p	μm	Length* at puberty	283	291	275	This study
L_i	μm	Ultimate length*	283	291	275	This study
W_d^i	μg	Ultimate dry weight	3.23	3.23	3.22	Goodman (1980); This study
J_o^p	nL h ⁻¹ μg ⁻¹	Ultimate dry-weight-specific O ₂ consumption	10.3	10.1	9.66	Herman and Heip (1983)
R_i	# d ⁻¹	Ultimate reproduction rate	7.57	8.20	7.97	Koch et al. (2017)

* Square root of the top view area as a length measure to scale with the cube root of structural volume.

highest food concentrations. While at this point we lack a physiological interpretation for the observed disparity, it may as well be a result of a non-ideal test design chosen by Koch et al. (2017). As the experiments were semi-static and different vessel sizes were used for the incubation of developing nauplii and copepodites versus reproducing females, slight deviations in the actual food availability per animal may also have caused some variation in the data. Overall, the functional response as outlined in the methods subsection 2.3.8 proved a simple, yet powerful one-parameter model extension to capture food dependency in the life history of *N. spinipes*.

3.4. Model comparison

While the overall data fit of the abp model was slightly better compared to the sbp model in view of the goodness of fit criteria, the difference seems not clear enough to deem sbp an implausible model representation of *N. spinipes*'s life history. A slightly different weighting of the data sets used in the parameter estimation might just as well have led to a better fit in sbp. In addition, abp and sbp are not the only models that can be thought of to describe the copepod life history. Jager et al. (2017) applied a switch in the feeding rate during one of the early copepodite stages to describe the growth of the copepod *Calanus finmarchicus*. Moreover, if metabolic acceleration actually occurs, it may also be limited to just a fraction of the copepod life history, e.g., the naupliar development. Additional data on *N. spinipes*'s life history could help to further investigate the occurrence of acceleration in this species. Data on the feeding rate over time, for example, can be used to approximate the assimilation rate which follows rather differentiable

trajectories in abp versus sbp (see Fig. 4). Data on the dry weight over time or the respiration rate over time might also be useful as long as the temporal resolution of measurements is high enough. However, due to the small size of *N. spinipes*, especially at the early life stages, such data are very difficult to obtain.

A first evaluation of metabolic acceleration in an evolutionary context was given only recently by Kooijman (2014). The author argued that metabolic acceleration is likely to have evolved at least five times independently in the animal kingdom. While all insects appear to accelerate, crustaceans are divided into non-accelerating taxa like water fleas and others that undergo different manifestations of type M metabolic acceleration. Due to the limited number of crustaceans in the AmP collection at this moment, an evolutionary classification of accelerating versus non-accelerating species remains speculative. High quality entries of other copepods and closely related taxa in the future, however, may enable better inter-species inferences.

3.5. Model applications

The main purpose of this study was to parametrize a DEB model for *N. spinipes* that can be used to investigate toxicity data in respect of possible physiological modes of action of chemical substances, here interpreted as alterations in DEB parameters (Jager et al., 2004). As both the abp and the sbp model for *N. spinipes* fit the observed life history data relatively well, at this point we cannot recommend one model over the other. We expect both models to reproduce toxic effect data equally well in most cases. If in special cases one model outperforms the other in explaining an observed dose-response relationship, this can bring important insights into the DEB rules of *N. spinipes* and help select the most accurate model.

Another useful feature of a DEB model is the scalability from the individual to a population under dynamic environmental conditions. While the individual is the key unit of the DEB theory, individual-based models can be used to simulate complex population dynamics that emerge from interactions between the individuals (Martin et al., 2012). Two individual level traits that can strongly influence the growth rate of a population are the development time until puberty and the reproduction rate per female (Hansen et al., 1999). In the abp model for *N. spinipes*, puberty is reached a little earlier and females have a slightly higher reproduction rate compared to the sbp model. Hence, a population model based on the abp model will predict a higher population growth rate under favorable environmental conditions. If food is a limiting factor, however, the food ingestion per animal gains importance. While initially abp animals have a lower assimilation rate compared to sbp, the opposite is the case at the adult stage as is illustrated in Fig. 4. Hence, a population dominated by adult animals would suffer more from intraspecific food competition in abp versus sbp. Despite these slight deviations, we hypothesize both models to

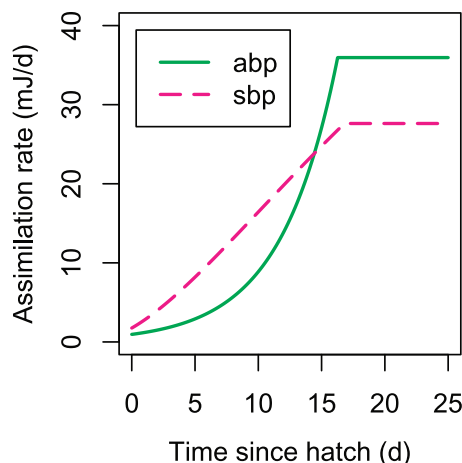


Fig. 4. Predicted assimilation rate over time in the typified models abp and sbp. Simulations were carried out at 22 °C with *ad libitum* food intake ($f = 1$).

perform very similarly in most individual-to-population extrapolations.

3.6. Add-my-Pet entry

With a steadily increasing number of entries during the past few years, the AmP collection has evolved to a rich data base that can be used, among other things, to explore patterns in DEB parameters throughout the animal kingdom. The Matlab package AmPtool which was released just recently facilitates data exploration in the AmP collection considerably (Marques et al., 2018). However, inter-specific parameter comparisons should be conducted with caution as the mathematical implications of certain parameters vary among the different typified models. The present study clearly demonstrates how different model types with different parameter values can lead to very similar results in their respective model output. Along with this study, an entry was made to the AmP collection with the abp model for *N. spinipes*. The abp model was chosen mainly to allow for parameter comparability with other copepod entries most of which are abp models at this point in time.

4. Conclusions

With extensive life history data available for *N. spinipes* we were able to parametrize the two typified DEB models abp (with metabolic acceleration) and sbp (with standard growth) for this species. The square root of the top view area proved to be a suitable length measure for length-to-volume conversions in *N. spinipes* as a species with changing body proportions. We suggest that the possibility of morphological changes should always be considered carefully in the process of DEB model parametrization since improper length-to-volume extrapolations can, for example, falsely indicate metabolic acceleration. The available experimental data were not strong enough to prove or reject the occurrence of metabolic acceleration in *N. spinipes* as both models showed almost equally good data fits. For most model applications, however, we hypothesize a similar performance of both models. More detailed data on the life history of *N. spinipes* and other closely related species in the future are needed to reveal the most accurate model.

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